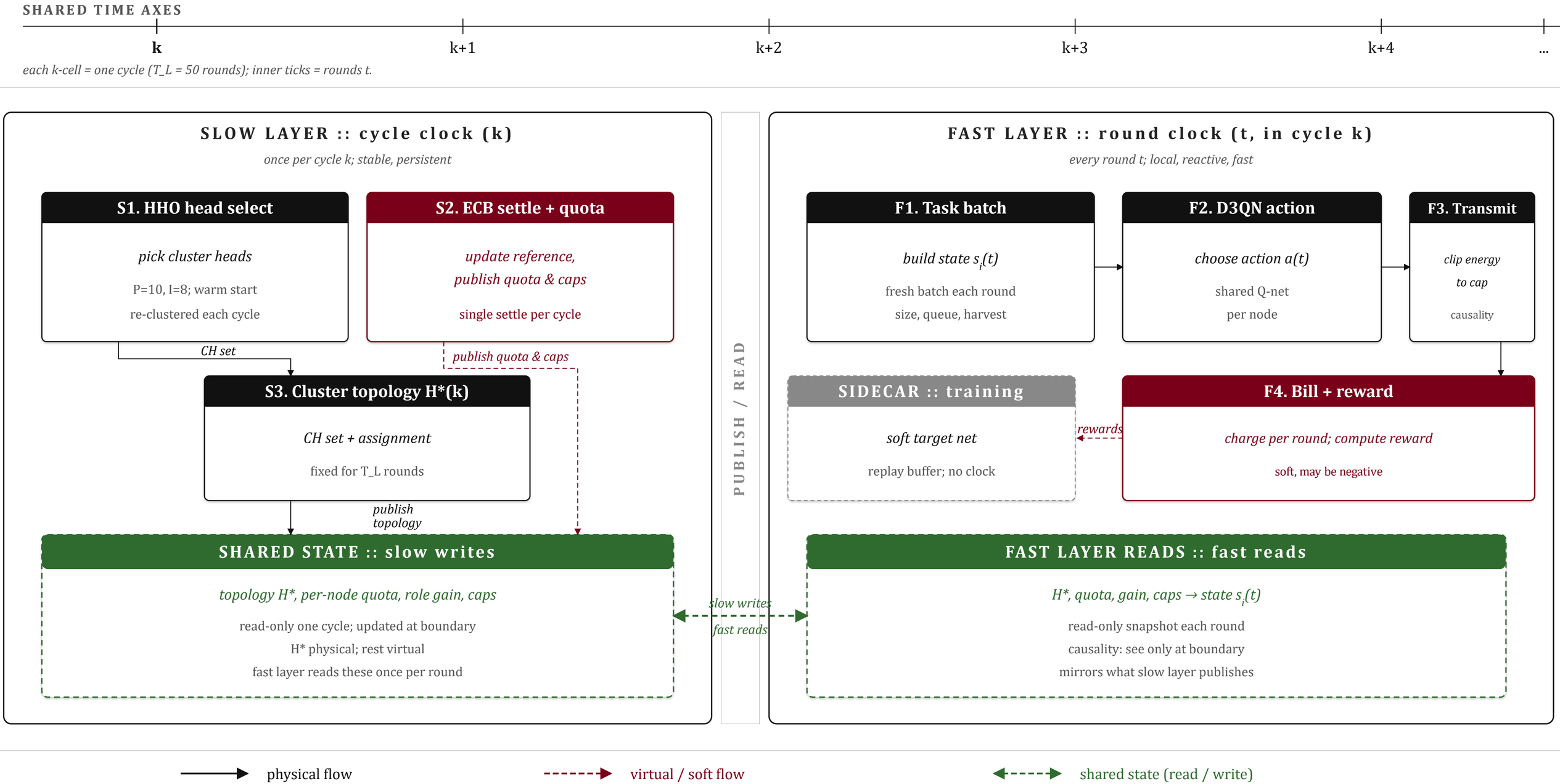

SHARED TIME AXES
k
k+1
k+2
k+3
k+4
...
each k-cell = one cycle (T_L = 50 rounds); inner ticks = rounds t.
SLOW LAYER :: cycle clock (k)
once per cycle k; stable, persistent
S1. HHO head select
pick cluster heads
P=10, I=8; warm start
re-clustered each cycle
S2. ECB settle + quota
update reference,
publish quota & caps
single settle per cycle
CH set
publish quota & caps
S3. Cluster topology H*(k)
CH set + assignment
fixed for T_L rounds
publish
topology
SHARED STATE :: slow writes
topology H*, per-node quota, role gain, caps
read-only one cycle; updated at boundary
H* physical; rest virtual
fast layer reads these once per round
PUBLISH / READ
slow writes
fast reads
FAST LAYER :: round clock (t, in cycle k)
every round t; local, reactive, fast
F1. Task batch
build state s_i(t)
fresh batch each round
size, queue, harvest
F2. D3QN action
choose action a(t)
shared Q-net
per node
F3. Transmit
clip energy
to cap
causality
SIDECAR :: training
soft target net
replay buffer; no clock
rewards
F4. Bill + reward
charge per round; compute reward
soft, may be negative
FAST LAYER READS :: fast reads
H*, quota, gain, caps → state s_i(t)
read-only snapshot each round
causality: see only at boundary
mirrors what slow layer publishes
physical flow
virtual / soft flow
shared state (read / write)